

INTERNAL USE ONLY

Implied Volatility Options Pricing Playbook

IV Rank, IV Percentile, expected move, and TradingView regime signals

Option Price = Intrinsic Value + Time Value

IV is the market's price of uncertainty, expressed through option premium.

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How to use this playbook

Use this as a desk reference for deciding whether option premium is relatively cheap or expensive, selecting a volatility-aware structure, and implementing the included TradingView indicator. It is designed for internal education and process consistency; it is not a trade recommendation engine.

Scope and governance

- Audience: traders, analysts, and risk reviewers working with listed options.
- Primary decision: determine whether the market is charging too much or too little for future movement.
- Required overlay: review catalyst timing, option-chain liquidity, bid/ask width, position limits, and defined-risk guidelines before execution.
- Data note: true contract-level IV should be checked in the option chain. The TradingView script can use an external volatility symbol or an historical-volatility proxy, but the proxy is not true IV.

Executive Summary

Implied volatility is a useful pricing barometer because it converts the market's expectation of future uncertainty into option premium. In practice, IV helps answer the desk's core options question: **are we paying too much, receiving enough, or passing on volatility risk?**

Option Price = Intrinsic Value + Time Value

IV mainly changes the time value component. Vega measures how much theoretical option price changes for a 1 percentage point change in IV. [3]

Concept	Operational read	Desk implication
High IV	The market is pricing larger expected movement and richer premium. [1]	Premium-selling structures may be favored, but tail risk and event risk must be controlled.
Low IV	The market is pricing calmer movement and cheaper optionality.	Premium-buying or debit structures may be favored when there is a catalyst or clear directional thesis.
IV Rank	Where current volatility sits between its own 1-year high and low.	Fast gauge of cheap vs. expensive premium relative to that product's history.
IV Percentile	Share of historical days with lower IV than today.	Better confirmation that current IV is unusual rather than distorted by one spike.
Expected move	Approximate range implied by IV and days to expiration.	Useful for setting strikes, judging break-even, and assessing whether the premium makes sense.

Bottom line

Options are not priced only on direction. They are priced on uncertainty, time, and probability. IV is the cleanest market-facing gauge of how expensive that uncertainty has become.

1. Why IV Is the Options Pricing Barometer

Implied volatility reflects the market's expectation for how much the underlying asset may fluctuate over a given period. Higher IV generally means higher expected movement and more expensive options; lower IV generally means cheaper options. [1]

The key operating principle is that premium is a price for uncertainty. When the market expects earnings surprises, macro events, product news, litigation outcomes, or broad risk-off behavior, buyers bid for optionality and IV can rise. After the event passes, IV can contract even if the trader was directionally correct.

Common beginner failure mode: direction right, volatility wrong

A long option can lose value after an event because the implied-volatility component collapses. This is often called IV crush. The trader may correctly forecast direction and still overpay for the option.

Question	Use IV to answer	Why it matters
Am I overpaying for movement?	Compare current IV to IV Rank and IV Percentile.	A good directional idea can still be unattractive if premium is too rich.
Is this an event-volatility trade?	Compare IV before and after catalysts such as earnings or scheduled data.	Event premium can collapse quickly once uncertainty is resolved.
Should I buy or sell premium?	Map IV regime to structure type: debit/long premium in low IV, credit/short premium in high IV.	Structure selection determines whether volatility expansion or contraction helps the position.
What range is being priced?	Estimate the IV-based expected move.	Helps select strikes and judge whether break-even is realistic.

2. Core Metrics

Current IV is useful, but it is not enough. A 45% IV may be high for a mega-cap utility-like stock, normal for a high-beta growth name, and low for a biotech before binary news. The desk should normalize IV against the product's own history.

2.1 IV Rank

$$\text{IV Rank} = (\text{Current IV} - \text{1Y IV Low}) / (\text{1Y IV High} - \text{1Y IV Low}) \times 100$$

Example: current IV 40%, 1-year low 20%, 1-year high 60% -> IV Rank = 50.

IV Rank band	Interpretation	Typical posture
0-20	Premium is relatively cheap versus its own history.	Consider long premium, debit spreads, or structures that benefit from volatility expansion.
20-50	Normal / mid-regime.	Prioritize directional edge, liquidity, and catalyst timing.
50-80	Premium is relatively expensive.	Consider defined-risk premium selling when risk/reward and event calendar are acceptable.
80-100	Extreme regime.	Use tighter risk controls; avoid assuming high IV cannot go higher.

2.2 IV Percentile

IV Percentile asks: what percentage of historical days had IV lower than today's IV? If IV Percentile is 82, current IV is higher than it was on 82% of days in the lookback window.

Metric	Best use	Weakness
IV Rank	Fast read of where current IV sits between historical high and low.	One unusual spike can distort the high/low range.
IV Percentile	More stable confirmation that current IV is actually unusual.	Can be slower to react to fast-changing regimes.
Desk rule Use IV Rank for speed and IV Percentile for confirmation. When both agree, the volatility regime is cleaner. When they diverge, look for one-off historical spikes, upcoming events, or data-source issues.		

3. Regime-Based Strategy Framework

The structure should align with the volatility thesis. Buying an option is partly buying volatility; selling an option is partly selling volatility. The same directional view can require different structures depending on IV regime.

IV condition	Directional / range view	Structures to evaluate	Primary risk
Low IV	Bullish	Long call, call debit spread	No catalyst; time decay exceeds move.
Low IV	Bearish	Long put, put debit spread	No catalyst; time decay exceeds move.
Low IV	Large move, uncertain direction	Long straddle, long strangle	Need enough movement before theta decay.
High IV	Bullish	Put credit spread	Gap down or volatility expands further.
High IV	Bearish	Call credit spread	Gap up or volatility expands further.
High IV	Neutral / range-bound	Iron condor, iron butterfly	Breakout beyond short strikes.
Extreme IV	Event uncertainty	Defined-risk only; consider smaller size	Tail move and assignment/early-exercise complexity.

Metric movement	Low-IV zone	High-IV zone
Vol rising	BUY VOL setup: cheap premium beginning to expand.	Risk-on event regime: do not sell premium blindly; confirm catalyst and tail risk.
Vol falling	Stay selective: premium cheap but getting cheaper.	SELL PREMIUM setup: rich premium beginning to contract.

Important: signal does not equal order

The included TradingView signals are regime flags, not buy/sell instructions. A trade still requires thesis, liquidity, event review, risk limit, and exit plan.

4. Expected Move and Risk Controls

Expected move converts annualized IV into an approximate move over the target holding period or expiration window. It is a rough range estimate, not a forecast or guarantee.

$$\text{Expected Move \%} = \text{IV} \times \sqrt{\text{Days to Expiration} / 252}$$

For crypto or 24/7 markets, use 365 annualization days. For US equities, 252 trading days is a common approximation.

Input	Example
Underlying price	\$100
IV	40%
DTE	30 days
Expected Move %	$40\% \times \sqrt{30 / 252} = 13.8\%$
Expected Move \$	\$13.80
Implied range	\$86.20 to \$113.80

4.1 Pre-Trade Quality Gate

Gate	Pass criteria
Volatility regime	IV Rank and/or IV Percentile supports the selected structure.
Catalyst calendar	Earnings, macro data, product event, FDA/legal decision, or ex-dividend date reviewed.
Liquidity	Tight bid/ask spread, adequate volume/open interest, executable contract size.
Risk definition	Max loss, max gain, break-even, and exit triggers are documented.
Position sizing	Risk fits desk limits; no unapproved naked short optionality.
Post-trade plan	Review date, adjustment trigger, and close/roll criteria are specified before entry.

Risk note

Short premium can have attractive probability characteristics but poor tail characteristics. Defined-risk spreads should be the default unless the desk has explicit approval, margin capacity, and documented assignment procedures.

5. TradingView Implementation Workflow

TradingView's options chain provides contract-level information such as strike, expiration, premiums, Greeks, and implied volatility; use the chain as the final source for trade-specific option data. [2] The included Pine Script indicator is a regime tool that can use either an external volatility symbol or an HV proxy.

Step	Action	Output
1	Open TradingView Pine Editor and paste the companion script.	Indicator available on chart.
2	Select data mode: external IV/volatility symbol when available; otherwise HV proxy.	Raw volatility series populated.
3	Set lookback to 252 for equity products, or adapt for desk-specific conventions.	IV Rank and IV Percentile normalized to the selected history.
4	Set thresholds: low 20, high 50, extreme 80 as defaults.	Regime zones visible on the indicator pane.
5	Set DTE for expected move and 252/365 annualization.	Approximate expected-move range appears in the dashboard.
6	Create alerts for Buy Premium Setup, Sell Premium Setup, and Extreme IV Regime.	Volatility regime changes routed into monitoring workflow.
7	Cross-check the exact target option in the options chain.	Trade-specific IV, Greeks, bid/ask, volume, and open interest reviewed.

5.1 Dashboard Field Guide

Dashboard field	Meaning	How to use
Data Mode	External IV/vol symbol or HV proxy.	Do not confuse HV proxy with true implied volatility.
Raw Vol	The underlying volatility input used by the indicator.	Check whether it is a percent-like series and whether the source is valid.
IV Rank	High/low normalized volatility reading.	Fast cheap/rich gauge.
IV Percentile	Percent of historical readings below current volatility.	Confirms whether current vol is unusually high/low.
Regime	Low, neutral, high, or extreme.	Use for structure selection.
Bias	Plain-English regime interpretation.	Use as a prompt for review, not an order signal.
Expected Move	Approximate price range over selected DTE.	Use for strike selection and break-even context.

6. Operating Procedures

This section turns the framework into a repeatable desk routine. The goal is to standardize how volatility is checked before any options trade is proposed.

Routine	Procedure
Daily scan	Review watchlist for IV Rank/Percentile extremes. Flag low-IV candidates with upcoming catalysts and high-IV candidates with potential post-event contraction.
Pre-trade memo	Record IV Rank, IV Percentile, DTE, expected move, event calendar, structure, max risk, and exit plan.

Routine	Procedure
Risk review	Confirm that the proposed structure benefits from the intended volatility path and does not introduce unapproved uncovered short optionality.
Trade management	Re-check IV after large price moves, after events, and when the position reaches predefined profit/loss or DTE thresholds.
Post-trade review	Classify result by direction, volatility forecast, timing, and execution quality. Capture whether IV expansion/contraction helped or hurt.

6.1 Common Exceptions

Exception	Why it matters	Control
Binary event	IV can stay elevated or expand further into the event.	Reduce size, define risk, and document event timing.
Volatility spike	IV Rank can look extreme but still rise if market stress accelerates.	Avoid selling premium solely because rank is high.
One-off historical spike	IV Rank may look artificially low for months after an extreme high.	Check IV Percentile and chart the raw vol series.
Skew / term structure	One headline IV reading does not capture strike-specific or expiration-specific pricing.	Review the target chain and compare nearby expirations.
Illiquid chain	Wide spreads can erase theoretical edge.	Avoid or reduce size unless execution quality is acceptable.

7. Indicator Limitations

The TradingView indicator is designed for regime awareness, not valuation certainty. It is intentionally simple so the desk can see when options are likely cheap/rich and then move to the option chain for trade-specific pricing.

- External IV symbol mode depends on the availability and quality of the selected volatility series.
- HV proxy mode uses realized historical volatility. It is useful context, but it is not implied volatility.
- One headline volatility input does not capture skew, term structure, liquidity, dividends, borrow constraints, or early-exercise considerations.
- `request.security()` can fetch data from another symbol/timeframe/context in Pine Script; use it carefully and avoid lookahead bias. [4]
- Pine Script evolves. This playbook uses Pine Script v6, which TradingView release notes identify as the version for future Pine updates as of the cited documentation. [5]

Desk control

The final trade ticket should always reference the actual option chain for the selected expiration and strike. The indicator can flag a regime, but the chain defines the price.

Appendix A. TradingView Pine Script v6 Indicator

The companion .pine file is included separately for clean copy/paste. The listing below is line-numbered for review and internal commentary.

```

001 | //@version=6
002 | indicator("IV Rank + IV Percentile Regime Signals", overlay=false, max_labels_count=500)
003 |
004 | //-----
005 | // Inputs
006 | //-----
007 | useExternalIV = input.bool(false, "Use external IV / volatility symbol?")
008 | ivSymbol      = input.symbol("CBOE:VIX", "External IV symbol")
009 | ivTimeframe   = input.timeframe("D", "IV data timeframe")
010 |
011 | hvLen         = input.int(20, "HV proxy length", minval=2)
012 | lookback      = input.int(252, "IV Rank / Percentile lookback", minval=30)
013 |
014 | lowZone       = input.float(20.0, "Low volatility threshold", minval=0, maxval=100)
015 | highZone      = input.float(50.0, "High volatility threshold", minval=0, maxval=100)
016 | extremeZone   = input.float(80.0, "Extreme volatility threshold", minval=0, maxval=100)
017 |
018 | signalMetric  = input.string("IV Rank", "Signal metric", options=["IV Rank", "IV Percentile"])
019 | dte           = input.int(30, "Expected move DTE", minval=1)
020 | annualDays    = input.string("252", "Annualization days", options=["252", "365"])
021 |
022 | showRawVol    = input.bool(true, "Show raw IV / HV line?")
023 | showSignals   = input.bool(true, "Show setup signals?")
024 | showTable     = input.bool(true, "Show dashboard?")
025 |
026 | //-----
027 | // External volatility data
028 | //-----
029 | externalVol = request.security(
030 |     ivSymbol,
031 |     ivTimeframe,
032 |     close,
033 |     gaps=barmerge.gaps_off,
034 |     lookahead=barmerge.lookahead_off,
035 |     ignore_invalid_symbol=true
036 | )
037 |
038 | //-----
039 | // Historical volatility proxy
040 | //-----
041 | days = annualDays == "365" ? 365.0 : 252.0
042 |
043 | logReturn = math.log(close / close[1])

```

```

044 | hvProxy   = ta.stdev(logReturn, hvLen) * math.sqrt(days) * 100.0
045 |
046 | //-----
047 | // Choose data source
048 | //-----
049 | hasExternal = useExternalIV and not na(externalVol)
050 | rawVol      = hasExternal ? externalVol : hvProxy
051 |
052 | modeText = hasExternal ? "External IV / Vol Symbol" : "HV Proxy"
053 |
054 | //-----
055 | // IV Rank calculation
056 | //-----
057 | volLow  = ta.lowest(rawVol, lookback)
058 | volHigh = ta.highest(rawVol, lookback)
059 |
060 | ivRankRaw = volHigh != volLow ? ((rawVol - volLow) / (volHigh - volLow)) * 100.0 : na
061 | ivRank    = math.min(math.max(ivRankRaw, 0.0), 100.0)
062 |
063 | //-----
064 | // IV Percentile calculation
065 | //-----
066 | ivPercentile = ta.percentrank(rawVol, lookback)
067 |
068 | //-----
069 | // Selected metric
070 | //-----
071 | metric = signalMetric == "IV Rank" ? ivRank : ivPercentile
072 |
073 | //-----
074 | // Regime logic
075 | //-----
076 | lowVol      = metric <= lowZone
077 | highVol     = metric >= highZone
078 | extremeVol  = metric >= extremeZone
079 |
080 | volRising   = metric > metric[1]
081 | volFalling  = metric < metric[1]
082 |
083 | // These are not buy/sell orders.
084 | // They are volatility regime signals.
085 | buyPremiumSetup = lowVol and volRising
086 | sellPremiumSetup = highVol and volFalling
087 |
088 | //-----
089 | // Expected move
090 | //-----
091 | expectedMovePct = rawVol * math.sqrt(dte / days)

```

```

092 | expectedMoveAbs = close * expectedMovePct / 100.0
093 |
094 | upperEM = close + expectedMoveAbs
095 | lowerEM = close - expectedMoveAbs
096 |
097 | //-----
098 | // Plots
099 | //-----
100 | metricColor =
101 |     extremeVol ? color.red :
102 |     highVol    ? color.orange :
103 |     lowVol     ? color.lime :
104 |     color.yellow
105 |
106 | plot(metric, "Selected IV Metric", color=metricColor, linewidth=2)
107 | plot(showRawVol ? rawVol : na, "Raw IV / HV %", color=color.new(color.blue, 20), linewidth=1)
108 |
109 | hline(0, "0", color=color.new(color.gray, 80))
110 | hline(lowZone, "Low Vol Zone", color=color.new(color.lime, 20), linestyle=hline.style_dashed)
111 | hline(highZone, "High Vol Zone", color=color.new(color.orange, 20), linestyle=hline.style_dashed)
112 | hline(extremeZone, "Extreme Vol Zone", color=color.new(color.red, 20), linestyle=hline.style_dashed)
113 | hline(100, "100", color=color.new(color.gray, 80))
114 |
115 | bgcolor(
116 |     extremeVol ? color.new(color.red, 88) :
117 |     highVol    ? color.new(color.orange, 90) :
118 |     lowVol     ? color.new(color.green, 90) :
119 |     na
120 | )
121 |
122 | //-----
123 | // Signal markers
124 | //-----
125 | plotshape(
126 |     showSignals and buyPremiumSetup,
127 |     title="Buy Premium Setup",
128 |     text="BUY\nVOL",
129 |     style=shape.triangleup,
130 |     location=location.bottom,
131 |     color=color.lime,
132 |     textcolor=color.black,
133 |     size=size.tiny
134 | )
135 |
136 | plotshape(
137 |     showSignals and sellPremiumSetup,
138 |     title="Sell Premium Setup",
139 |     text="SELL\nPREM",

```

```

140 |         style=shape.triangledown,
141 |         location=location.top,
142 |         color=color.red,
143 |         textcolor=color.white,
144 |         size=size.tiny
145 |     )
146 |
147 | //-----
148 | // Dashboard table
149 | //-----
150 | var table dash = table.new(position.top_right, 2, 9, border_width=1)
151 |
152 | if showTable and barstate.islast
153 |     regimeText =
154 |         extremeVol ? "Extreme IV" :
155 |         highVol    ? "High IV" :
156 |         lowVol     ? "Low IV" :
157 |         "Neutral IV"
158 |
159 |     biasText =
160 |         buyPremiumSetup ? "Buying premium improving" :
161 |         sellPremiumSetup ? "Selling premium improving" :
162 |         lowVol          ? "Premium cheap, wait for expansion" :
163 |         highVol         ? "Premium rich, watch for crush" :
164 |         "Neutral"
165 |
166 |     table.cell(dash, 0, 0, "Data Mode", text_color=color.white, bgcolor=color.black)
167 |     table.cell(dash, 1, 0, modeText, text_color=color.white, bgcolor=color.black)
168 |
169 |     table.cell(dash, 0, 1, "Raw Vol")
170 |     table.cell(dash, 1, 1, str.toString(rawVol, "#.##") + "%")
171 |
172 |     table.cell(dash, 0, 2, "IV Rank")
173 |     table.cell(dash, 1, 2, str.toString(ivRank, "#.##"))
174 |
175 |     table.cell(dash, 0, 3, "IV Percentile")
176 |     table.cell(dash, 1, 3, str.toString(ivPercentile, "#.##"))
177 |
178 |     table.cell(dash, 0, 4, "Signal Metric")
179 |     table.cell(dash, 1, 4, signalMetric)
180 |
181 |     table.cell(dash, 0, 5, "Regime")
182 |     table.cell(dash, 1, 5, regimeText)
183 |
184 |     table.cell(dash, 0, 6, "Bias")
185 |     table.cell(dash, 1, 6, biasText)
186 |
187 |     table.cell(dash, 0, 7, "Expected Move")

```

```
188 |         table.cell(dash, 1, 7, "+/-" + str.toString(expectedMoveAbs, "###") + " / " + str.toString(expectedMovePct, "###") + "%")
189 |
190 |         table.cell(dash, 0, 8, "EM Range")
191 |         table.cell(dash, 1, 8, str.toString(lowerEM, "###") + " - " + str.toString(upperEM, "###"))
192 |
193 | //-----
194 | // Alerts
195 | //-----
196 | alertcondition(buyPremiumSetup, "Buy Premium Setup", "IV is low and starting to expand. Consider long premium / debit structures.")
197 | alertcondition(sellPremiumSetup, "Sell Premium Setup", "IV is high and starting to contract. Consider defined-risk premium selling structures.")
198 | alertcondition(extremeVol, "Extreme IV Regime", "IV Rank / Percentile is in the extreme zone.")
```

References and Source Notes

Sources were used to anchor definitions and platform-specific implementation notes. Accessed April 19, 2026.

- [1] TradingView Support. *Implied volatility*. Notes that implied volatility reflects market expectations of future fluctuation and that higher IV generally leads to more expensive options. <https://www.tradingview.com/support/solutions/43000735608-implied-volatility/>
- [2] TradingView Support. *Options chain overview*. Lists option-chain fields including strike, expiration, premiums, Greeks, and implied volatility. <https://www.tradingview.com/support/solutions/43000760837-options-chain-overview/>
- [3] TradingView Support. *Vega option greek*. Defines Vega as the theoretical option price sensitivity to a 1% change in implied volatility. <https://www.tradingview.com/support/solutions/43000708853-vega-option-greek/>
- [4] TradingView Pine Script Documentation. *Other timeframes and data*. Explains that Pine scripts can request data from other symbols, timeframes, and contexts. <https://www.tradingview.com/pine-script-docs/concepts/other-timeframes-and-data/>
- [5] TradingView Pine Script Documentation. *Release notes*. Includes the Pine Script v6 release note and recommendation to convert existing scripts to v6 for new features. <https://www.tradingview.com/pine-script-docs/release-notes/>

Internal disclaimer

This document is for internal educational and process-design purposes only. It does not constitute investment advice, a recommendation to buy or sell securities, or a complete risk-management framework. Options involve substantial risk and are not suitable for all investors.

Version	Date	Change
v1.0	April 19, 2026	Initial playbook draft with IV framework, TradingView implementation workflow, and Pine Script appendix.